

Based on 48 sources

Operator_name	Formal_definition	Anti_structure_mapping	Anti_pattern_mapping
Boundary Operator (∂)	A mapping from an n-simplex to its boundary, specifically defined as the union of faces opposite to each vertex: $\partial\Delta_n = \bigcup_{0 \leq i \leq n} d_i(\Delta_n)$.	Interior mapping or the inclusion of an n-simplex into its larger cone ($\Delta_n \subseteq \Delta_{n+1}$).	Incorrectly assuming the boundary operator in a higher category must be invertible without an n-groupoid structure.
Join Operator (?)	An operation on simplicial sets S and S' defined as the union over all decompositions of a linearly ordered set J into disjoint subsets I and I' where $i < i'$ for all elements, effectively joining every vertex of S to every vertex of S'.	Disjoint union ($\amalg$) or fiber product ($\times_S$) where no connections exist between the sets.	Using the join operator to simply append data (concatenation) rather than establishing a topological 'link' between every element of the input sets.
Cayley Transforms (Isomorphic to S-Enriched Path Objects)	In the context of Higher Topos Theory, this relates to the construction of a path object P(U) for an S-enriched category U, involving functorial factorization of the diagonal map $A \rightarrow A \times A$.	Discrete mapping or identity functor where no 'path' or transformation exists between points.	Using path factorizations in prompts without ensuring the model maintains 'weak equivalence' (coherence) across the transformation.